ARIMAX Model for the Gross Internal Electrical Energy Consumption in Spain 1990 - 2020

Adrian Scholl[a,1] **and Ludvig Bergstrand**[a,1]

[a] *Universitat Politècnica de Catalunya (UPC), Faculty of Mathematics and Statistics (FME)*

Laboratory professors: Lesly Acosta, Laura Marquès

Contents

1 Introduction 1

2 A. Classic ARIMA Analysis 2

2.1 Identification . . . 2
Achieving Stationarity • Model Selection

2.2 Estimation . . . 4

2.3 Validation . . . 5
ACF and PACF Diagnostics • Residual Assumptions • Stationarity (Causality) and Invertibility

2.4 Prediction . . . 10

2.5 Forecasting . . . 10

3 B. ARIMAX Analysis 11

3.1 Outlier and calendar effects treatment . . . 11
Calendar effects • Outlier detection and adjustment

3.2 Validation . . . 12
ACF and PACF Diagnostics • Residual Assumptions • Stationarity (Causality) and Invertibility

3.3 Prediction . . . 16

3.4 Forecasting . . . 16

4 C. Comparison of ARIMA and ARIMAX 16

5 Conclusion 17

5.1 Key Findings . . . 17

5.2 Limitations and Outlook . . . 17

5.3 Final Remark . . . 17

6 Appendix 18

1. Introduction

Accurate modeling and forecasting of electricity consumption are crucial for efficient energy management and planning. Forecasts enable utilities and planners to balance supply and demand, plan infrastructure, and integrate renewable energy sources. As Jahanshahi *et al.*[1] note, the estimation of energy consumption is fundamental to economic planning and climate change mitigation. Time-series models are particularly effective in this context: they leverage historical consumption data to project future demand[2].

Electricity consumption time series typically exhibit pronounced seasonal patterns, long-term trends, and occasional irregular shocks. For example, in Spain (as in many temperate climates) consumption peaks in both summer and winter due to cooling and heating demands. These characteristics create modeling challenges: series with strong trend or seasonality require differencing or seasonal terms to achieve stationarity, and external effects (weather, economic cycles) may warrant exogenous regressors. Forecasting is thus complicated by a "complex interplay between long-term trends, seasonal fluctuations, and other unpredictable factors"[4]. Recent surveys emphasize that accounting explicitly for trend and seasonality is essential for accurate forecasts[3].

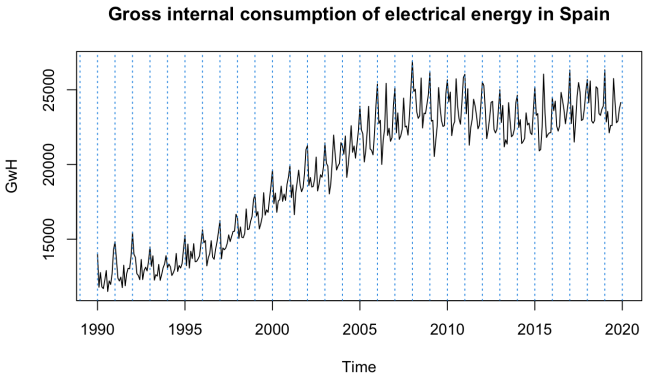


Figure 1. Gross internal consumption of electrical energy in Spain (monthly, GWh) from 1990 to 2020 (data from Banco de España).

Figure 1 shows the monthly series of gross internal electrical consumption in Spain over 1990–2020 with 360 monthly observations from Banco de España. An overall upward trend is evident, reflecting steadily growing demand. Superimposed on this trend are seasonal oscillations: consumption reaches peaks in summer and winter (likely driven by cooling and heating loads). Notably, the series exhibits a sharp decline around 2009, coinciding with the global financial crisis. This drop introduces a structural break: consumption falls sharply in 2009 and resumes growth thereafter. We treat this 2009 event as a potential outlier in our analysis.

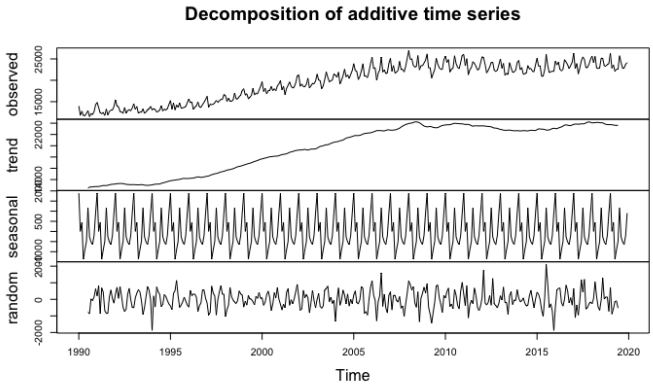


Figure 2. Additive decomposition of the electricity consumption series into trend, seasonal, and residual components.

The seasonal decomposition in Figure 2 further clarifies these patterns. The trend component rises until about 2008 and then flattens, illustrating the structural break. The seasonal component shows a clear annual cycle with peaks in summer and winter, consistent with Fig. 1. The residual component is small except around 2009, where the crisis-induced drop appears as an outlier. In summary, the series comprises a smooth upward trend, a dominant annual seasonal cycle, and an isolated shock around 2009. These observations guide our modeling strategy.

To forecast consumption, we use SARIMA and SARIMAX models, which explicitly capture trend, seasonality, and exogenous influences. The SARIMA (Seasonal ARIMA) framework extends ARIMA by including seasonal differencing and terms to model recurring patterns[3]. SARIMAX (SARIMA with eXogenous variables) further incorporates external regressors into the ARIMA structure. Both approaches have proven effective in energy forecasting: for example, Alharbi and Csala report that SARIMAX "significantly improved forecasting accuracy" compared to simpler ARIMA-based models[2]. Similarly, Maleki *et al.* found that SARIMAX (with external inputs) "fared well" in predicting aggregated electricity load[4]. We will implement these models using standard Box–Jenkins identification and validation procedures on the 1990–2020 Spain series and evaluate their predictive performance.

The rest of this report is organized as follows. Section 2 presents additional exploratory insights to complement the introduction. Section 3 outlines the classical ARIMA modeling process, including identification, estimation, validation, prediction, and forecasting. Section 4 extends the analysis to ARIMAX models, incorporating outlier and calendar effects as exogenous variables. Section 5 compares the results of ARIMA and ARIMAX models and in Section 6 we summarize our findings and conclude the report.

2. A. Classic ARIMA Analysis

2.1. Identification

At first sight, the initial time series in Figure **??** shows a clear positive trend, signs of yearly seasonality and also a variance that seems to increase with increasing electricity consumption. As the ARIMA framework requires a stationary data generating process, we begin with checking for non-constant variance, non-constant mean and seasonality and eventually transforming our time series into a stationary process.

2.1.1. Achieving Stationarity

To confirm the **heterogeneity** we look at the yearly mean and variances in Figure 3 and the yearly boxplot in Figure 4, where each boxplot characterizes the distribution of the 12 monthly values per year.

The mean-variance plot reveals a clear, positive correlation between yearly means and variances. The yearly boxplots also show a tendency for stronger fluctuation at higher consumption levels. Together this confirms the suspicion of heterogeneity.

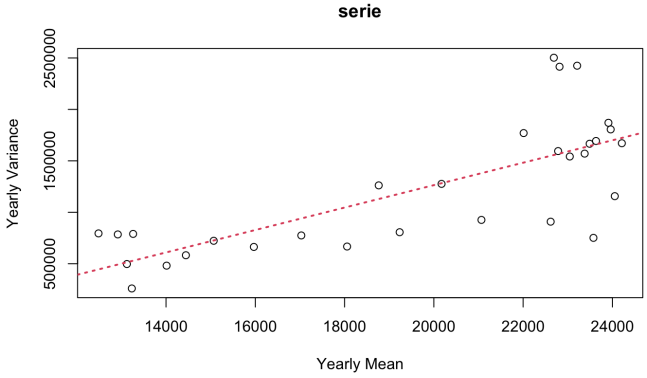


Figure 3. Yearly mean versus yearly variance of the initial timeseries.

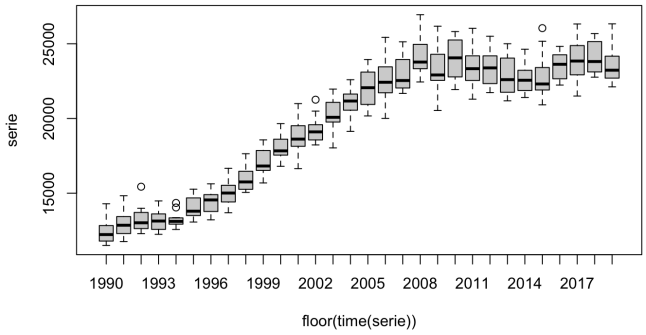


Figure 4. Yearly boxplots of the initial time series.

This type of heterogeneity can typically be fixed by applying a **log-transformation** to the initial time series. Thus, we apply the log-transformation to stabilize the variance and look at the plots again in the Figures 5 and 6.

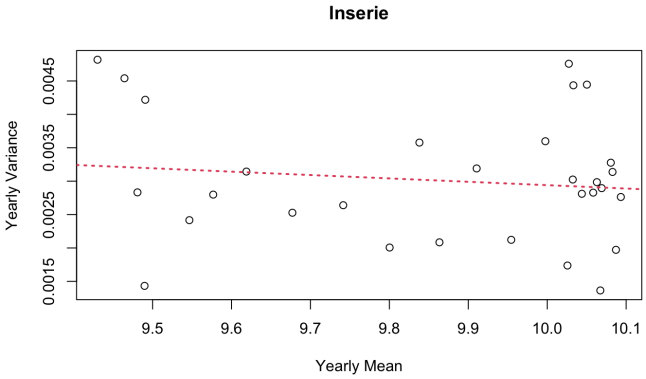


Figure 5. Yearly mean versus yearly variance of the log-transformed timeseries.

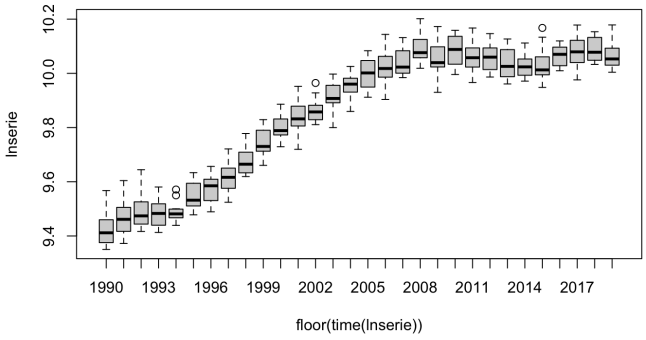


Figure 6. Yearly boxplots of the log-transformed time series.

After the log-transformation, there correlation between yearly means and variances negligibly small and the inter-quartile ranges in the boxplots appear much more stable. Thus, we conclude that the yearly variance is seemingly constant and the process is yearly homoscedastic now.

To confirm possible **yearly seasonality** we plot the decomposition of the log-transformed time series in Figure 7 and the monthplot in

Figure 8 showing the mean of the log-transformed electricity consumptions grouped by month.

The decomposition shows a systematic seasonal component. Specifically, we can observe in the monthplot that the electricity consumption is in the mean higher in the winter months January and December than in the autumn and spring months which could arise from a more intense heating behavior during these cold months. We see another seasonal increase in the consumption for July that could hypothetically be explained by the high number of tourists visiting Spain in this period or an increased usage of air conditioning facilities due to generally higher temperatures.

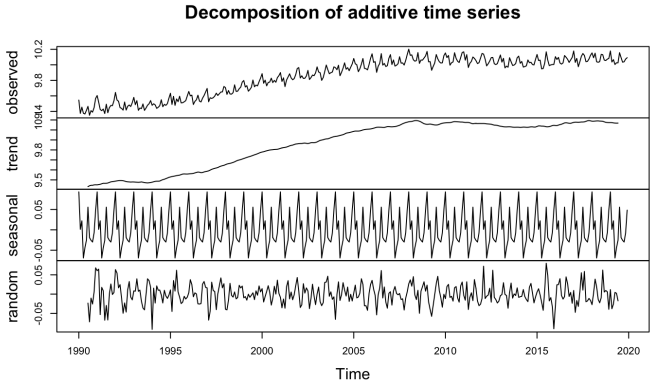


Figure 7. Decomposition of log-transformed time series.

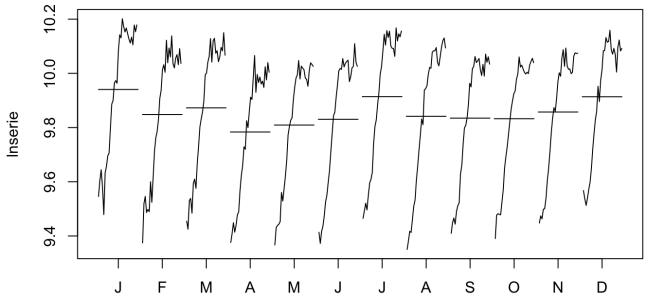


Figure 8. Monthplot of the log-transformed time series.

We decide to account for this seasonal trend by applying a seasonal difference with $s = 12$. Figure 9 confirms that the seasonal trend could be successfully removed and the monthly means are now constant. Furthermore, we observe that for $D = 1$ the variance is minimal and increases again for $D = \{2, 3\}$, as shown in Table 1. Thus, $D = 1$ is the right choice for deseasonalizing the time series.

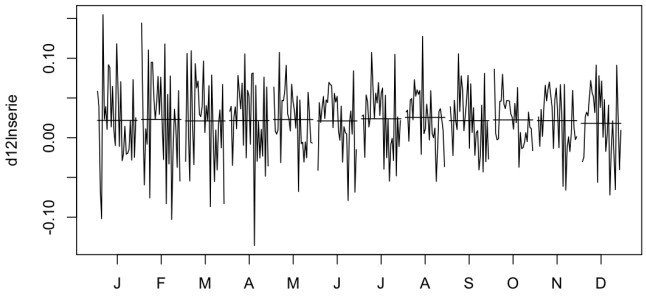


Figure 9. Monthplot of the log-transformed time series after seasonal differencing with $D = 1$.

In Figure 10 we see that the mean now seems to be nearly constant. It looks like the first order seasonal differencing might have also removed the general upward trend. To ensure that we do not need additional regular differencing, we compare the variance for applying $d = 1, 2, 3$ with the variance of the current process with $d = 0$. Table 1 shows that the variance increases slightly for $d = 1$ and even further for $d > 1$.

From this we follow that the time series $B^{12}\log(X_t)$ is indeed stationary and we can now use it to identify a suitable ARIMA model.

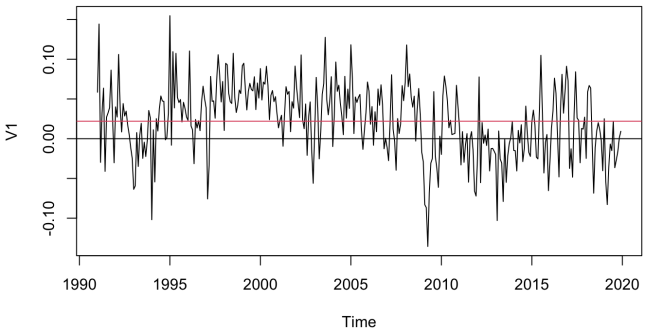


Figure 10. The log-transformed time series after seasonal differencing with $D = 1$.

Table 1. Variance of log-transformed series under seasonal and regular differencing.

Differencing Type	Order	Variance
Seasonal Differencing D on $\log(X_t)$	$D = 0$	0.05736
	$D = 1$	0.00203
	$D = 2$	0.00470
	$D = 3$	0.01546
Regular Differencing d on $B^{12}\log(X_t)$	$d = 0$	0.00203
	$d = 1$	0.00217
	$d = 2$	0.00598
	$d = 3$	0.01916

2.1.2. Model Selection

We now use the ACF and PACF plots in Figures 11 12 to identify possible ARIMA models: The ACF shows 10 significantly positive, fast decaying regular lags. The PACF shows a clearly significant first positive regular lag, followed by a second on barely significant positive

lag, a third insignificant lag and a fourth again barely significant positive lag. After that the following lags are insignificant again until we again get positive lags due to seasonal partial autocorrelation effects.

For the **regular ARIMA part** we follow that an AR with $p = 1$ or $p = 2$, is most likely with p definitively being $p \leq 4$. Also an MA process with up to $q = 10$ could be derived, which however seems highly overparametrized.

For the seasonal ARIMA part we observe the ACF and PACF at the seasonal lags (multiples of 12) marked in red. Both, the ACF and PACF show a fast decaying seasonal pattern. The ACF has a first significantly negative lag, then a smaller second and third significant lag. The fourth seasonal is only barely significantly positive, followed only by insignificant lags. The PACF shows a first and second clearly negative seasonal lag and a third barely significantly negative seasonal lag, followed by only insignificant lags. We follow for the **seasonal ARIMA part** that either a SMA(3)/SMA(4) process or a SAR(2)/SAR(3) process are most likely.

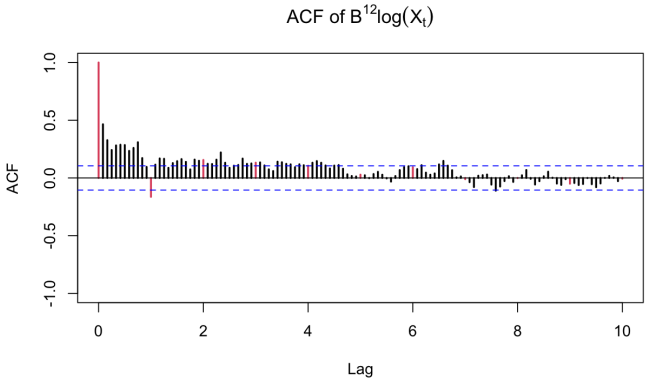


Figure 11. ACF of the stationary time series $B^{12}\log(X_t)$.

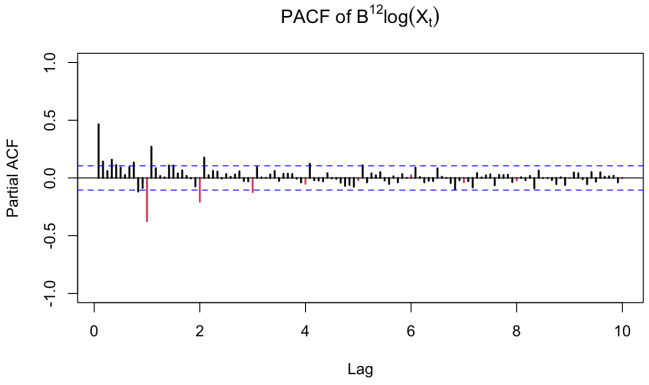


Figure 12. PACF of the stationary time series $B^{12}\log(X_t)$.

2.2. Estimation

After the identification of plausible parametrizations based on the ACF and PACF correlograms, we conduct a systematic **grid search** with a grid provided in Table 2 containing all regular and seasonal AR and MA lags up to the highest plausible lag derived from the ACF and PACF. We decide to include the intercept in all models as the true mean of the series $B^{12}\log(X_t)$ could be proven to be significantly different from 0 with a t-test with p-value $< 2.2\times10^{-16}$ for the sample mean of 0.02203 and thus carries important information about the level of the series. To prevent overparametrization we constrain the search space of possible models to $p + q + P + Q \leq 10$. Additionally, after the grid search we exclude all models with errors in the estimation process due to invalid parameter combinations and all models that yielded insignificant parameter estimates for p, q, P, Q.

Table 2. Grid of ARIMA model orders used for model selection

Component	Candidate values
Regular AR order (p)	0, 1, 2, 3, 4
Regular MA order (q)	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Seasonal AR order (P)	0, 1, 2, 3
Seasonal MA order (Q)	0, 1, 2, 3, 4

Table 3 reveals the ten best model from this grid search according to the AIC. It becomes immediately apparent that the ARIMA(2,0,1)(0,1,1) and ARIMA(1,0,2)(0,1,1) model are performing best with respect to AIC (-1431.480 and -1430.814) and also with respect to the stricter penalizing BIC (-1408.164 and -1407.497) since both need only 4 parameters. The third best model ARIMA(3,0,3)(0,1,1) yields a similar AIC of -1430.219, but since it is much more complex with 7 parameters the BIC is much worse with a value of -1395.244.

Therefore, we decide to select the models **ARIMA(2,0,1)(0,1,1)** and **ARIMA(1,0,2)(0,1,1)**.

Table 3. ARIMA Model Grid Search Results

Model	AIC	BIC	LogLik	Pms.
ARIMA(2,0,1)(0,1,1)	-1431.480	-1408.164	721.7401	4
ARIMA(1,0,2)(0,1,1)	-1430.814	-1407.497	721.4068	4
ARIMA(3,0,3)(0,1,1)	-1430.219	-1395.244	724.1094	7
ARIMA(4,0,3)(0,1,1)	-1425.461	-1386.600	722.7304	8
ARIMA(1,0,1)(0,1,1)	-1422.616	-1403.185	716.3080	3
ARIMA(1,0,2)(3,1,0)	-1417.572	-1386.483	716.7859	6
ARIMA(3,0,2)(3,1,0)	-1414.103	-1375.242	717.0514	8
ARIMA(3,0,3)(3,1,0)	-1412.359	-1369.612	717.1793	9
ARIMA(1,0,1)(3,1,0)	-1411.834	-1384.631	712.9170	5
ARIMA(2,0,1)(2,1,0)	-1407.573	-1380.370	710.7865	5

Table 4 presents the estimated coefficients, their standard errors, and the corresponding 95% confidence intervals for both models, assuming asymptotic normality of the estimators according to the Central Limit Theorem (CLT).

We observe that Model 1, we observe that the AR(1) and seasonal MA(1) coefficients (`ar1` and `sma1`) are statistically significant, as their 95% confidence intervals do not include zero. Similarly, in Model 2, the AR(1), MA(1), MA(2), and seasonal MA(1) terms are significant at the 5% level. The intercept estimate in both models is not significantly different from zero, and can therefore be neglected for practical purposes.

Table 4. SARIMA model coefficient estimates with standard deviations and 95% confidence intervals

ARIMA (2,0,1) (0,1,1)				
Param.	**Estimate**	**Std. Error**	**95% CI**	
ar1	1.2258	0.0687	1.0901	1.3615
ar2	-0.2298	0.0679	-0.3639	-0.0957
ma1	-0.8088	0.0384	-0.8849	-0.7327
sma1	-0.7892	0.0377	-0.8631	-0.7153
ARIMA (1,0,2) (0,1,1)				
Param.	**Estimate**	**Std. Error**	**95% CI**	
ar1	0.9945	0.0052	0.9843	1.0047
ma1	-0.5867	0.0518	-0.6880	-0.4854
ma2	-0.1582	0.0487	-0.2539	-0.0625
sma1	-0.7889	0.0378	-0.8633	-0.7145

2.3. Validation

The two elected models from the previous chapter are now validated by first comparing their sample ACF and PACF with their theoretical counterparts. In the following, the assumption of independently homoscedastic (normally) distributed residuals is verified using statistical tests and graphical validation. Finally, we investigate the stationarity and invertibility of both models.

2.3.1. ACF and PACF Diagnostics

For the first model **ARIMA(2,0,1)(0,1,1)**) we can see that the sample ACF plot resembles the theoretical ACF very accurately. The spikes for the MA(1) at lag 1 and for the SMA(1) at lag 12 are clearly visible. The decay for the AR(2) process is strongly reflected in the sample ACF.

The sample PACF also follows its theoretical counterpart closely with the first positive AR(2) lag clearly visible and the second lag barely visible, as it is also very small in the theoretical PACF. Also here the exponentially decaying behavior of the lags here are visible and support the modeling of the regular part as MA(1)/SMA(1) model.

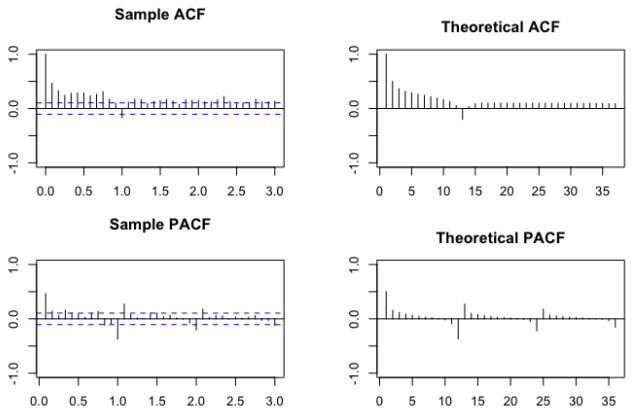


Figure 13. Theoretical and Sample ACF and PACF of ARIMA(2,0,1)(0,1,1)

The theoretic ACF and PACF of the second model **ARIMA(1,0,2)(0,1,1)** looks very similar to the ones of the first model. Thus, the sample ACF and PACF also support this model. The difference is that now the second regular AR lags should be smaller and the second regular MA lag larger.

As the second regular sample PACF lag is very small while the first two regular ACF lags are clearly significant, the sample ACF and PACF show slightly more evidence for the second model.

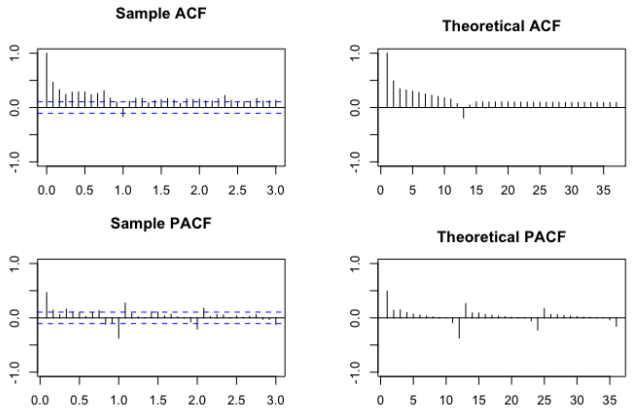


Figure 14. Theoretical and Sample ACF and PACF of ARIMA(1,0,2)(0,1,1)

2.3.2. Residual Assumptions

In this section, we assess whether the residuals of both models meet the assumption of white noise. This ensures that all meaningful structure in the time series is captured by the models. As a first overview we report the distributional overviews of the residuals shown in Figure 15 and Figure 16.

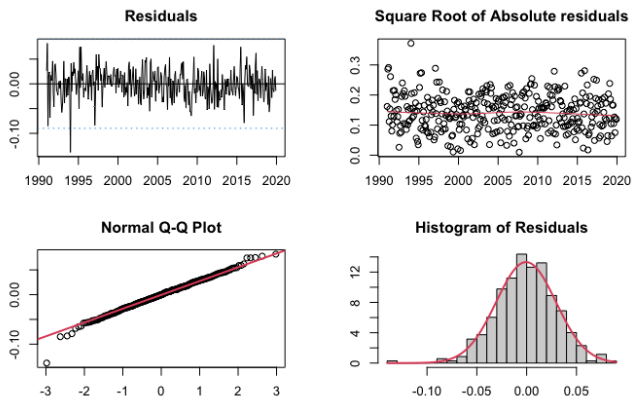


Figure 15. Residual Overview for ARIMA(2,0,1)(0,1,1)

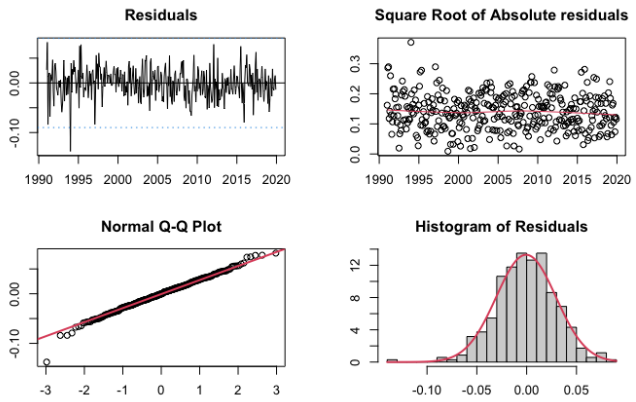


Figure 16. Residual Overview for ARIMA(1,0,2)(0,1,1)

To quantitatively confirm the assumption of white noise we conduct a in-depth analysis of the three distributional characteristics that it implies:

1. Zero Mean Residuals should be centered around zero. We investigate this by checking for symmetry around the zero line in the

residual plots shown in Figures 15 and 16. At first sight, both residual series seem to fluctuate around zero.

As quantitative confirmation, we report both residual sample means and conduct a **one-sample t-test** checking the null hypothesis of a zero mean, assuming normally distributed residuals.

Table 5. One-sample t-test results for zero-mean residuals

Statistic	ARIMA(2,0,1)(0,1,1)	ARIMA(1,0,2)(0,1,1)
Mean (Sd.)	−0.00059 (0.0300)	−0.00057 (0.0300)
t-Statistic	−0.3687	−0.3549
p-Value	0.7126	0.7229
95% CI	[−0.00375, 0.00257]	[−0.00373, 0.00259]

The results of the t-tests, displayed in Table 5, support the assumption that residuals are centered around zero. The sample means are very close to zero for both models, and the corresponding p-values are well above the significance threshold of $\alpha = 5\%$. Hence, we fail to reject the null hypothesis that the residual means are zero. Therefore, the assumption of zero-mean residuals is considered **fulfilled for both models**.

2. Homoskedasticity Residuals should share the same constant variance over time. We assess homoskedasticity both visually and quantitatively. Visually, we examine the ACF and PACF of the squared residuals in Figures 17 and 18. For both models, there are no significant autocorrelations, with the exception of a small spike around lag 36 months, which appears weak and not systematically relevant. This suggests no strong signs of heteroskedasticity.

Additionally, the plots of the square root of the absolute residuals in Figures 15 and 16 reveal no visible patterns or time-varying trends in the residual scale, further supporting the homoskedasticity assumption.

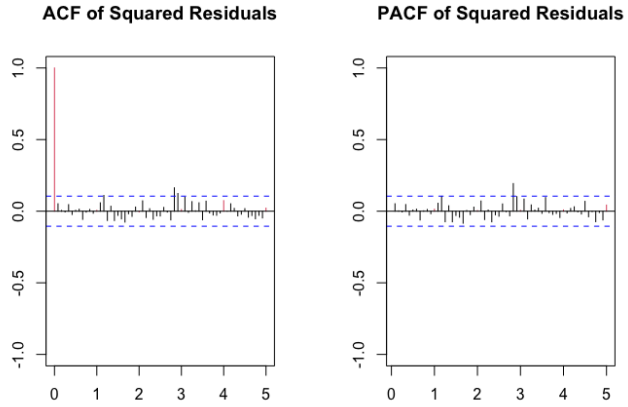


Figure 17. ACF and PACF of the Squared Residuals of ARIMA(2,0,1)(0,1,1)

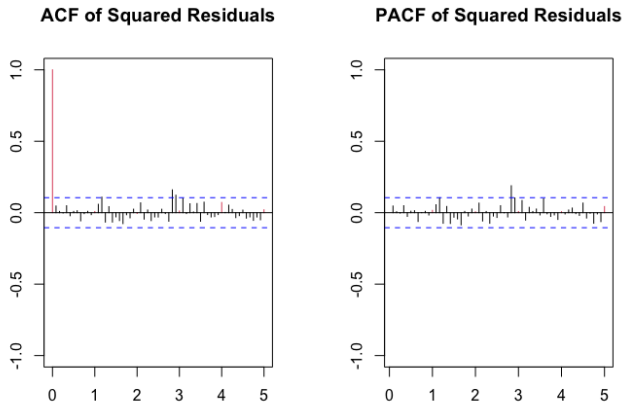


Figure 18. ACF and PACF of the Squared Residuals of ARIMA(1,0,2)(0,1,1)

To substantiate this visual assessment, we apply a formal statistical test:

Studentized Breusch-Pagan Test

The studentized Breusch–Pagan test examines whether the variance of the residuals remains constant over time. The null hypothesis (H_0) assumes homoskedasticity, meaning that the residuals exhibit constant variance. The alternative hypothesis (H_1) suggests heteroskedasticity—that the residual variance depends on the model's fitted values or other regressors. The test statistic is given by

$$LM = nR^2 \sim \chi^2_k,$$

where R^2 is obtained from an auxiliary regression of the squared residuals on the fitted values.

In the context of SARIMA models, the fitted values typically serve as the only regressor in the auxiliary regression, so the degrees of freedom k equals one. A high p-value indicates that the null hypothesis cannot be rejected, suggesting that the residual variance appears constant and there is no significant evidence of heteroskedasticity.

Table 6. Breusch–Pagan test for homoskedasticity

Statistic	ARIMA(2,0,1)(0,1,1)	ARIMA(1,0,2)(0,1,1)
BP statistic	0.0796	0.0814
Degrees of freedom	1	1
p-value	0.7779	0.7754

The studentized Breusch–Pagan test yields high p-values for both models ($p = 0.7779$ for ARIMA(2,0,1)(0,1,1) and $p = 0.7754$ for ARIMA(1,0,2)(0,1,1)), as reported in Table 6. Thus, the null hypothesis of constant variance cannot be rejected. Together with the visual analysis, this confirms that the **homoskedasticity assumption is sufficiently met for both SARIMA models.**

3. Independence Residuals should be independent and should not exhibit autocorrelation. We first take a look at the **ACF and PACF plots of the residuals** to visually inspect any signs of autocorrelation. For both the SARIMA(2,0,1)(0,1,1) model and the SARIMA(1,0,2)(0,1,1) model, the plots (Figure 19 and Figure 20) show that all autocorrelations lie within the 95% confidence bounds, indicating no significant lag structure beyond what is expected from white noise.

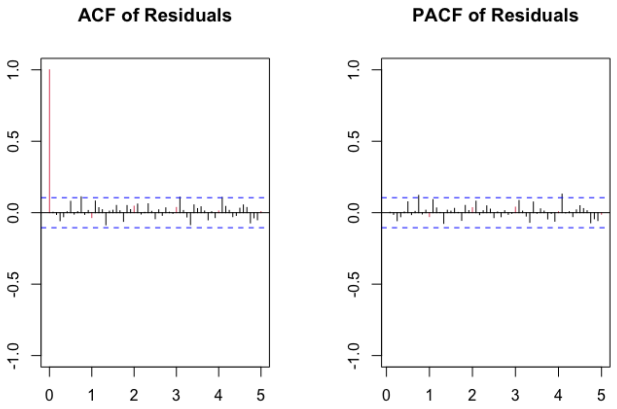


Figure 19. ACF and PACF plots of residuals for SARIMA(2,0,1)(0,1,1).

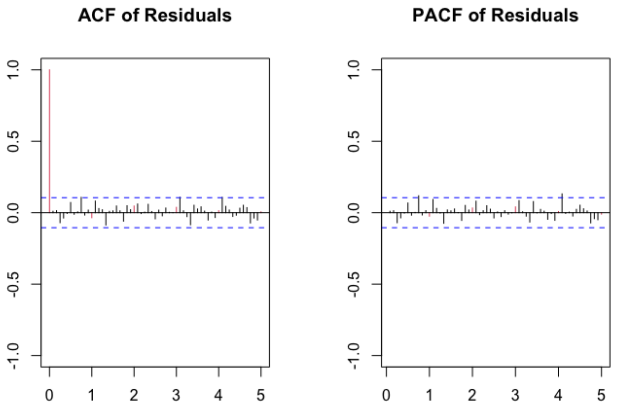


Figure 20. ACF and PACF plots of residuals for SARIMA(1,0,2)(0,1,1).

To quantitatively confirm that the residuals from the fitted models are free from autocorrelation, we apply both the **Durbin-Watson test** and the **Ljung-Box test**.

Durbin-Watson Test

The **Durbin-Watson (DW) test** is specifically designed to detect first-order autocorrelation in the residuals of regression-type models. Its test statistic is defined as

$$DW = \frac{\sum_{t=2}^{T}(e_t - e_{t-1})^2}{\sum_{t=1}^{T} e_t^2},$$

where e_t denotes the residual at time t. The test statistic ranges from 0 to 4, with a value close to 2 suggesting no first-order autocorrelation. Values significantly below 2 imply positive autocorrelation, while values above 2 imply negative autocorrelation.

Model	DW Statistic	p-value
ARIMA(2,0,1)(0,1,1)	2.0083	0.5093
ARIMA(1,0,2)(0,1,1)	1.9935	0.4544

Table 7. Durbin-Watson test results for both models. The DW statistics near 2 and the non-significant p-values indicate no first-order autocorrelation.

As shown in Table 7, both models yield DW statistics close to 2 with high p-values, providing no evidence against the null hypothesis of no autocorrelation.

Ljung-Box Test

The **Ljung-Box (LB) test** is a portmanteau test that checks whether a group of autocorrelations in the residuals up to lag h are jointly zero. The test statistic is defined as

$$Q = n(n+2)\sum_{k=1}^{h}\frac{\hat{\rho}_k^2}{n-k} \overset{H_0}{\sim} \chi^2_{h-m},$$

where n is the sample size and $\hat{\rho}_k$ is the sample autocorrelation at lag k and m the number of estimated parameters. Under the null hypothesis H_0 the residuals are not autocorrelated and the statistic follows a chi-squared distribution with $h - m$ degrees of freedom.

Model	Lag	Statistic	p-value
ARIMA(2,0,1)(0,1,1)	12	8.7751	0.7220
	24	18.9001	0.7573
	36	23.7706	0.9413
	48	35.4254	0.9110
ARIMA(1,0,2)(0,1,1)	12	9.0043	0.7026
	24	18.7847	0.7634
	36	23.5101	0.9459
	48	34.9026	0.9211

Table 8. Ljung-Box test results at multiple lags for both models. The high p-values indicate no significant autocorrelation in the residuals.

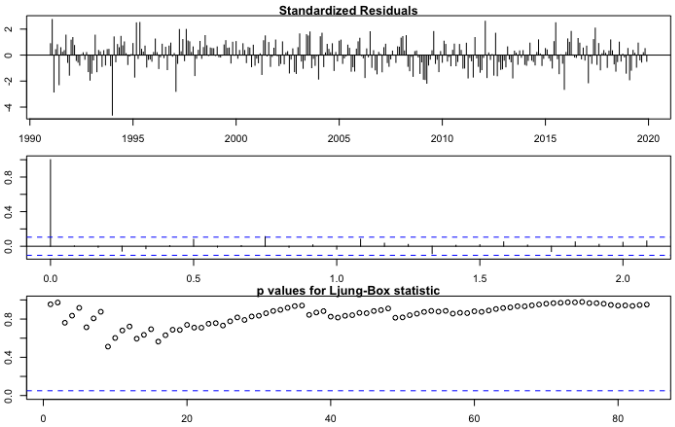


Figure 21. Standardized residuals and p-values for the Ljung-Box test up to $h = 84$ for ARIMA(2,0,1)(0,1,1).

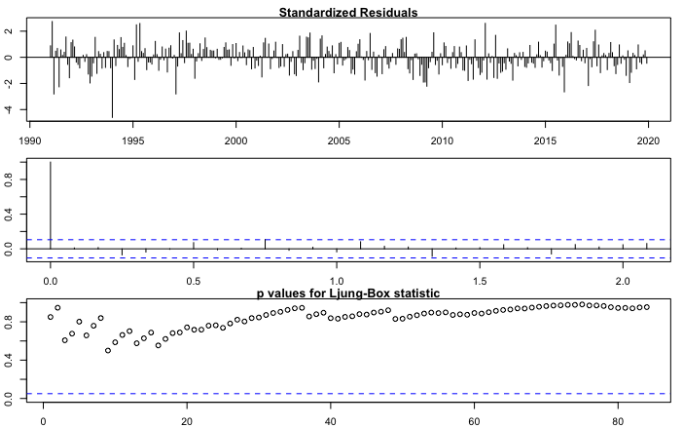


Figure 22. Standardized residuals and p-values for the Ljung-Box test up to $h = 84$ for ARIMA(1,0,2)(0,1,1).

The test results in Table 8 and Figure 22 and Figure 22 confirm that for both models, the residuals do not exhibit significant autocorrelation at any tested lag. All p-values are substantially above common significance thresholds.

In conclusion both visual and statistical assessments provide consistent evidence that the residuals from both SARIMA models are free from significant autocorrelation.

4. Normality Residuals should be approximately normally distributed if inference is needed. To begin, we visually assess the distribution of the residuals using QQ-plots and histograms in the residual overview displayed in Figure 15 and Figure 16.

Both QQ-plots indicate that the sample quantiles align well with the theoretical quantiles of the normal distribution, suggesting approximate normality. However, there are visible negative deviations in the lower tails, particularly with the final data point lying substantially above the QQ-line. The corresponding histograms confirm this impression: both residual distributions appear bell-shaped and symmetric, yet exhibit an increased left tail. These observations suggest that while normality is largely met, slight left-tailed deviations remain.

To quantitatively assess normality, we apply the **Shapiro-Wilk test**, **Anderson-Darling test**, and **Jarque-Bera test** to the standardized residuals of both models.

Shapiro-Wilk Test

The Shapiro-Wilk test evaluates the null hypothesis that a sample $x_1, \dots, x_n$ comes from a normally distributed population. The test statistic is

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2},$$

where $x_{(i)}$ denotes the ordered sample values and a_i are constants computed from the covariance matrix of the order statistics. Under the null hypothesis of normality, W is close to 1. Smaller values and small p-values indicate departure from normality.

Anderson-Darling Test

The Anderson-Darling test is a modification of the Kolmogorov-Smirnov test that gives more weight to the tails. It tests the null hypothesis that a sample comes from a specified distribution—in this case, the normal distribution. The test statistic is

$$A^2 = -n - \frac{1}{n}\sum_{i=1}^{n}(2i-1)\left[\log F(x_{(i)}) + \log(1 - F(x_{(n+1-i)}))\right],$$

where F is the cumulative distribution function (CDF) of the normal distribution. Under the null hypothesis, A^2 is small.

Jarque-Bera Test

The Jarque-Bera test is based on the skewness and kurtosis of the sample. It tests the null hypothesis that the data has skewness 0 and excess kurtosis 0, like for a normal distribution. The test statistic is

$$JB = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \overset{H_0}{\sim} \chi^2_{df=2},$$

where S is the sample skewness and K is the sample kurtosis. Large values of JB and small p-values suggest non-normality.

Table 9. Normality test results for residuals of both SARIMA models

Test	Model	Statistic	p-value
Shapiro-Wilk	ARIMA(2,0,1)(0,1,1)	0.9910	0.0327
	ARIMA(1,0,2)(0,1,1)	0.9910	0.0318
Anderson-Darling	ARIMA(2,0,1)(0,1,1)	0.2351	0.7905
	ARIMA(1,0,2)(0,1,1)	0.2661	0.6889
Jarque-Bera	ARIMA(2,0,1)(0,1,1)	18.1910	0.0001
	ARIMA(1,0,2)(0,1,1)	18.2060	0.0001

As seen in Table 9, the results are mixed. The Shapiro-Wilk tests yield p-values slightly below the conventional significance threshold of 0.05, suggesting mild deviations from normality. The Anderson-Darling tests, in contrast, report high p-values, indicating no evidence against normality. Meanwhile, the Jarque-Bera tests clearly reject the null hypothesis of normality, with extremely low p-values. Given the visual evidence and the mixed test outcomes, we conclude that the residuals deviate slightly from normality—especially in the tails—but are overall sufficiently close for inference purposes.

Combining the findings of the autocorrelation and normality analyses, we conclude that the residuals of both models satisfy the white noise assumption: they are uncorrelated and homoskedastic. Regarding Gaussian noise, although some tests indicate mild deviations from normality—particularly in the tails—the residuals exhibit approximate normality overall. Thus, for practical purposes of inference and forecasting, the **white noise and approximate Gaussian assumptions are reasonably met**.

2.3.3. Stationarity (Causality) and Invertibility

A SARIMA model is defined as

$$\Phi(B^s)\phi(B)(1-B)^d(1-B^s)^D y_t = \Theta(B^s)\theta(B)\varepsilon_t,$$

where $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$ and $\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$ are the regular AR and MA polynomials and $\Phi(B^s) = 1 - \Phi_1 B^s - \cdots - \Phi_P B^{Ps}$ and $\Theta(B^s) = 1 + \Theta_1 B^s + \cdots + \Theta_Q B^{Qs}$ are the seasonal AR and MA polynomials, respectively.

The SARIMA model is **stationary** or **causal** if all roots of the combined autoregressive polynomial $\Phi(B^s)\phi(B)$ lie outside the unit circle, meaning that their moduli are strictly greater than 1. This condition ensures that the process (after differencing) has a constant mean and variance over time. Tables 21 and 22 in the Appendix show the AR roots of both models.

For the ARIMA(2,0,1)(0,1,1) model the moduli of the AR roots are 1.005305 and 4.327857, both greater than 1. The AR root of ARIMA(1,0,2)(0,1,1) has a modulus of 1.005522, also greater than 1. Hence, both models satisfy the condition for stationarity.

We also plot the π-**weights** for both models that represent the AR(∞) form of the SARIMA models in Figure 23 and Figure 24. The π-weights show how past values influence current values for each lag and should exhibit a fast decay if the model is stationary.

The π-weights of ARIMA(2,0,1)(0,1,1) decay rapidly from 0.42 (lag 1) to about 0.02 (lag 10), indicating weak long-term dependence and supporting stationarity. For the second model the π-weights also decay steadily from 0.41 to around 0.02 by lag 10, with small oscillations beyond lag 12. This consistent decay also indicates stationarity while showing the effects of the seasonality in the series.

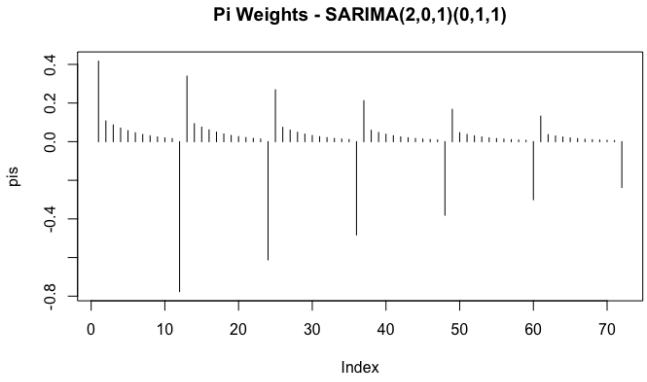


Figure 23. π-weights of ARIMA(2,0,1)(0,1,1).

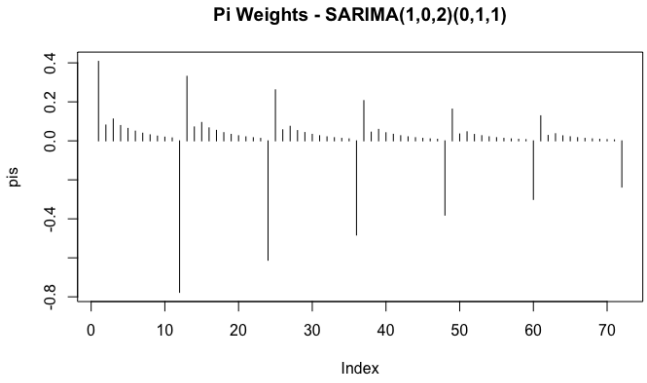


Figure 24. π-weights of ARIMA(1,0,2)(0,1,1).

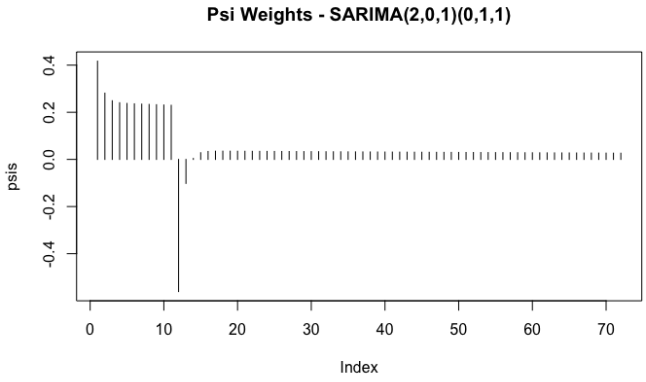


Figure 25. ψ-weights of ARIMA(2,0,1)(0,1,1).

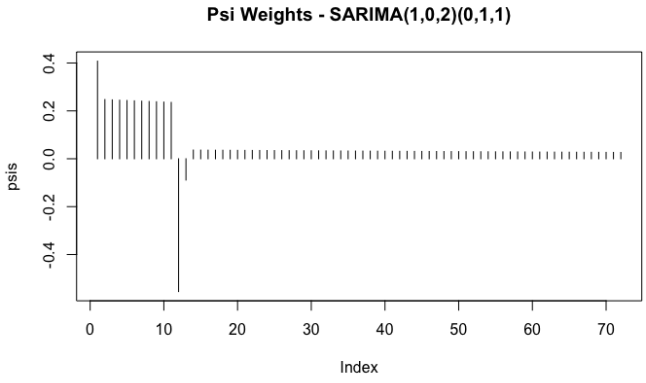


Figure 26. ψ-weights of ARIMA(1,0,2)(0,1,1).

Based on the modulus of AR and MA roots, as well as the behavior of π and ψ weights shown in Figures 23–26, **both models fulfill the conditions for stationarity (causality) and invertibility**.

The model is **invertible** if all roots of the combined moving average polynomial $\Theta(B^s)\theta(B)$ lie outside the unit circle. This guarantees a unique representation as a convergent infinite-order AR(∞) process. Tables 21 and 22 in the Appendix show the MA roots of both models.

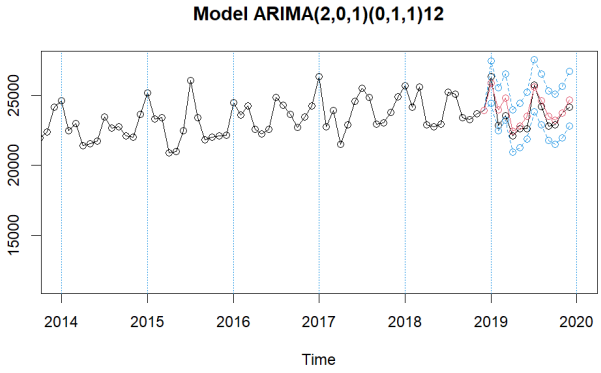
The ARIMA(2,0,1)(0,1,1) has only MA root moduli that are greater than 1 indicating invertibility. For ARIMA(1,0,2)(0,1,1) all MA root moduli are also greater than 1, ranging from 1.019957 to 4.979622, confirming invertibility.

The ψ-**weights** plotted in Figure 25 and 26 represent the MA(∞) form of the process and decay over time if the model is invertible.

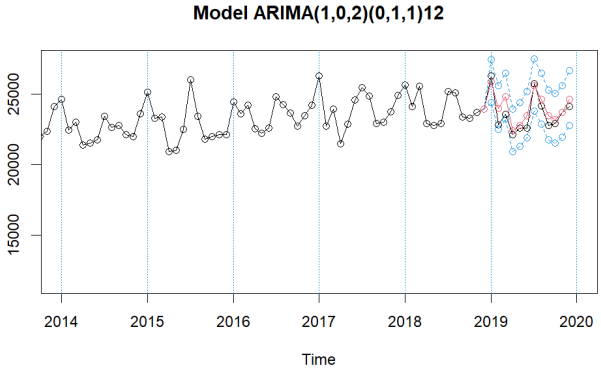
The ψ-weights of ARIMA(2,0,1)(0,1,1) start at 0.42 for lag 1 and then decrease abruptly to approximately 0.035 by lag 20. From this decline we read that the model is invertible. For ARIMA(1,0,2)(0,1,1) the ψ-weights show a decrease with a similar pattern, beginning at 0.41 and dropping to around 0.035 by lag 20. A minor dip is observed at lag 12, but the overall downward trend confirms that this model is likewise invertible.

2.4. Prediction

To assess the predictive performance and stability of the selected models, we reserved the last 12 months of the series for out-of-sample validation. Using the estimated SARIMA models, we generated one-step-ahead forecasts and compared them to the actual observed values. The predictions can be observed in Figures 27.



(a) Prediction with ARIMA(2,0,1)$(0,1,1)_{12}$



(b) Prediction with ARIMA(1,0,2)$(0,1,1)_{12}$

Figure 27. Stability and one-year predictions for the two selected ARIMA models, showing the observed series, predicted values, and 95% confidence intervals.

Error metrics

Let y_t be the observed log-transformed and seasonally adjusted consumption at time t, and $\hat{y}_t$ the predicted value from the model. We computed the following forecast accuracy measures for each model:

- **Root Mean Squared Error (RMSE)**:

$$\text{RMSE} = \sqrt{\frac{1}{h}\sum_{t=1}^{h}(y_t - \hat{y}_t)^2}$$

- **Mean Absolute Error (MAE)**:

$$\text{MAE} = \frac{1}{h}\sum_{t=1}^{h}|y_t - \hat{y}_t|$$

- **Root Mean Squared Percentage Error (RMSPE)**:

$$\text{RMSPE} = \sqrt{\frac{1}{h}\sum_{t=1}^{h}\left(\frac{y_t - \hat{y}_t}{y_t}\right)^2}$$

- **Mean Absolute Percentage Error (MAPE)**:

$$\text{MAPE} = \frac{1}{h}\sum_{t=1}^{h}\left|\frac{y_t - \hat{y}_t}{y_t}\right|$$

These metrics provide insight into both the accuracy and precision of the predictions. Lower values indicate better model performance. The mean length of the 95% prediction interval was also used to assess the forecast uncertainty. The error metrics can be observed in Table 10.

Model	RMSE	MAE	RMSPE	MAPE
ARIMA(2,0,1)$(0,1,1)_{12}$	646.03	526.66	0.02779	0.02253
ARIMA(1,0,2)$(0,1,1)_{12}$	646.67	524.91	0.02783	0.02246

Table 10. Forecast error metrics for the two selected ARIMA models over the 2019 validation period.

Stability

The stability of the two selected ARIMA models was evaluated by the forecasts for the 12-month period following December 2018 and comparing them to the observed values. As shown in fig 27, both ARIMA(1,0,2)$(0,1,1)_{12}$ and ARIMA(2,0,1)$(0,1,1)_{12}$ produced forecasts that remain closely aligned with the seasonal structure and level of the historical data. In both cases, the predicted values lie well within the 95% confidence intervals, and there is no visible sign of systematic bias, trend misspecification, or structural instability.

The models maintained consistent behavior across the forecast horizon, and their error metrics (RMSE, MAE, RMSPE, MAPE) remained low and stable throughout. These findings suggest that both models exhibit good short-term forecast stability and are adequately capturing the underlying dynamics of the series. Slight deviations between predicted and observed values can be attributed to inherent seasonality and noise, rather than model misfit.

2.5. Forecasting

After validating the models, we used the full series to re-estimate the final versions of the two best-performing SARIMA models. Using these models, we produced forecasts for the subsequent 12 months beyond the available data.

Forecast computation

Let $\hat{y}_{T+h}$ denote the h-step-ahead forecast at time T, the last observed point in the training data. Forecasts were computed recursively using the fitted model coefficients and previous observations:

$$\hat{y}_{T+h} = \mu + \sum_{i=1}^{p}\phi_i\hat{y}_{T+h-i} + \sum_{j=1}^{q}\theta_j\hat{\varepsilon}_{T+h-j} + \text{seasonal terms}$$

Where:

- ϕ_i: AR coefficients
- θ_j: MA coefficients
- $\hat{\varepsilon}_{T+h-j}$: residuals from previous periods

The prediction intervals were computed assuming normally distributed forecast errors:

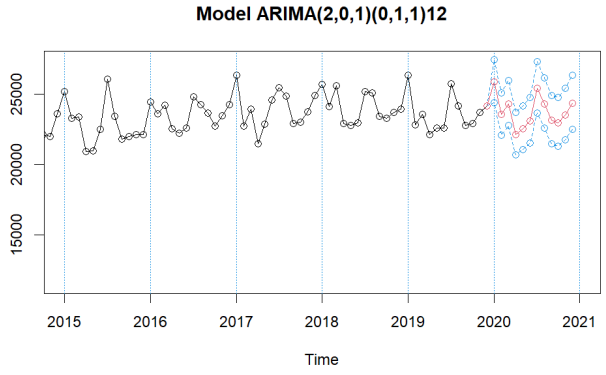
$$\hat{y}_{T+h} \pm z_{\alpha/2} \cdot \hat{\sigma}_h$$

where $\hat{\sigma}_h$ is the standard error of the h-step-ahead forecast and $z_{\alpha/2}$ corresponds to the 97.5th percentile of the standard normal distribution (for 95% intervals).

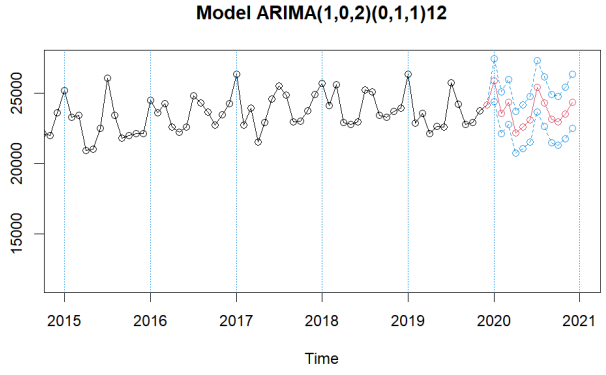
Since the model was estimated on the logarithmic scale to stabilize variance, the forecasts and confidence intervals were back-transformed using the exponential function:

$$\hat{Y}_{T+h} = \exp(\hat{y}_{T+h})$$
$$\text{CI}_{\text{lower}} = \exp(\hat{y}_{T+h} - 1.96 \cdot \hat{\sigma}_h)$$
$$\text{CI}_{\text{upper}} = \exp(\hat{y}_{T+h} + 1.96 \cdot \hat{\sigma}_h)$$

The forecasts can be observed in Figure 29.



(a) Forecast with ARIMA(2,0,1)$(0,1,1)_{12}$



(b) Forecast with ARIMA(1,0,2)$(0,1,1)_{12}$

Figure 28. One-year forecast for the two selected ARIMA models, showing the observed series, predicted values, and 95% confidence intervals.

Interpretation

The forecasted values and their confidence intervals provide a projection of future electricity consumption. The width of the confidence intervals reflects the model uncertainty. Although both ARIMA(1,0,2)$(0,1,1)_{12}$ and ARIMA(2,0,1)$(0,1,1)_{12}$ demonstrate acceptable forecasting performance, several limitations must be acknowledged. Firstly, the models rely solely on the internal structure of the time series, without incorporating potentially influential external variables such as temperature, economic activity, or regulatory changes. Secondly, they assume a fixed seasonal pattern and do not accommodate evolving or nonlinear seasonality, which may be relevant in electricity consumption data. Additionally, abrupt structural changes—such as sudden drops or spikes—are not explicitly modeled and may degrade forecast accuracy. The prediction intervals remain relatively wide, reflecting forecast uncertainty that could potentially be reduced with exogenous information. Lastly, while the models were validated over a one-year forecast horizon, their long-term forecasting performance remains untested.

Let Y_t denote the observed (log-transformed) electricity consumption at time t, and let ε_t denote a white noise innovation. The backshift operator B is defined by $BY_t = Y_{t-1}$.

The first model, ARIMA(1,0,2)$(0,1,1)_{12}$, is expressed as:

$$(1 - \phi_1 B)(1 - B^{12})Y_t = (1 + \theta_1 B + \theta_2 B^2)(1 + \Theta_1 B^{12})\varepsilon_t,$$

where ϕ_1 is the non-seasonal AR(1) coefficient, θ_1 and θ_2 are the non-seasonal MA(1) and MA(2) coefficients, and Θ_1 is the seasonal MA(1) coefficient with seasonal period 12.

The second model, ARIMA(2,0,1)$(0,1,1)_{12}$, is written as:

$$(1 - \phi_1 B - \phi_2 B^2)(1 - B^{12})Y_t = (1 + \theta_1 B)(1 + \Theta_1 B^{12})\varepsilon_t,$$

where ϕ_1 and ϕ_2 are the non-seasonal AR(1) and AR(2) coefficients, θ_1 is the non-seasonal MA(1) coefficient, and Θ_1 is again the seasonal MA(1) coefficient.

3. B. ARIMAX Analysis

3.1. Outlier and calendar effects treatment

This section extends the ARIMA analysis by incorporating calendar-related covariates and accounting for structural outliers. These adjustments aim to improve model accuracy and interpretability by capturing systematic effects not explained by the standard seasonal ARIMA structure.

3.1.1. Calendar effects

Electricity consumption is influenced by the number of working days in a month and the timing of holidays such as Easter. To quantify these effects, we constructed the following external regressors:

- **Trading day effect (TD)**: Reflects the deviation in the number of working days (weekdays) in each month compared to a reference configuration (e.g., 5 weekdays / 2 weekend days).
- **Easter effect (EAST)**: Captures shifts in Easter holidays that can affect consumption, especially when Easter moves between March and April.

These variables were generated using pre-supplied functions (`Wtrad()` and `Weaster()`) and incorporated into a SARIMAX model via the `xreg` argument.

$$Y_t = \mu + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i} + \sum_{j=1}^{Q} \Theta_j \varepsilon_{t-12j} + \beta_1 \cdot \text{TD}_t + \beta_2 \cdot \text{EAST}_t + \varepsilon_t$$

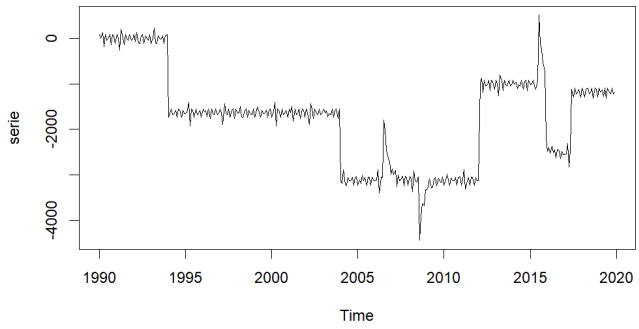
where Y_t denotes the observed electricity consumption, and ε_t is the error term. The estimated coefficients for the calendar effects were interpreted, and their significance was evaluated using confidence intervals. In both models, the trading day effect and Easter effect were found to be statistically significant. The model taking into account both trading day effects and easter effects obtains a smaller AIC than using them seperatly.

3.1.2. Outlier detection and adjustment

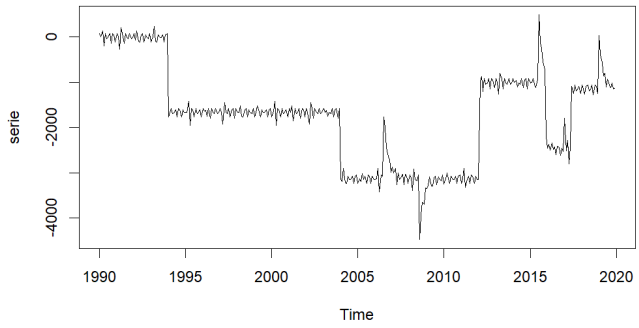
Time series data often include irregular events (e.g., energy crises, holidays, reporting issues) that manifest as structural outliers. We used the implementation (`outdetec()`) with critical value $c = 2.8$ and `LS=TRUE` to ensure additional control over linearization and visual inspection. Outliers were then removed by constructing a linearized version of the series with these anomalous values replaced or corrected. The outliers can be observed in Table 11 and the diffrence in fig 29b.

Obs	Type	W_coeff	ABS L-ratio	Date	% Effect
49	LS	-1632.61	3.55	Ene 1994	0
169	LS	-1486.24	3.47	Ene 2004	0
199	TC	1440.13	3.71	Jul 2006	∞
224	TC	-1202.92	3.25	Ago 2008	0
266	LS	2115.01	4.52	Feb 2012	∞
307	TC	1432.41	3.60	Jul 2015	∞
312	LS	-1535.03	3.50	Dic 2015	0
330	LS	1362.08	3.21	Jun 2017	∞

Table 11. Detected outliers for time series



(a) Difference from time serie with adjusted calendar effects for ARIMA(2,0,1)(0,1,1)$_{12}$ model.



(b) Difference from time serie with adjusted calendar effects for ARIMA(1,0,2)(0,1,1)$_{12}$ model.

Figure 29. Difference of time series and models adjusted for calendar effects.

Interpretation of detected outliers

The outliers include both level shifts (LS) and temporary changes (TC), each of which impacts the interpretation and modeling of the time series.

Five level shift (LS) outliers were detected in January 1994, January 2004, February 2012, December 2015 and June 2017. These correspond to abrupt and permanent changes in the level of electricity consumption. The LS in February 2012 and June 2017 shows a large positive shift of approximately 2115 GWh and 1362 GWh, suggesting a structural increase in consumption. In contrast, the LS detected in January 1994, January 2004, December 2015 reflect significant downward shifts of -1632 GWh, -1486 GWh and -1535 GWh respectively, possibly due to policy changes, economic downturns, or unusual weather patterns in those periods.

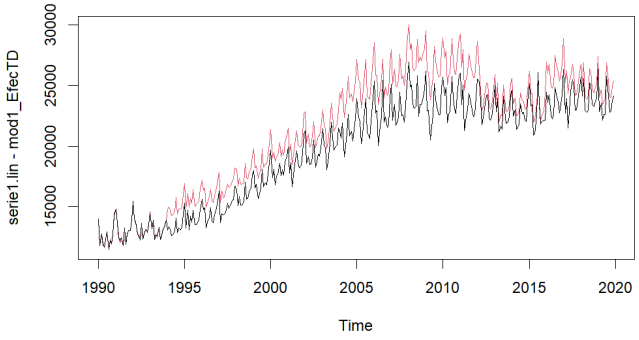
Temporary change (TC) outliers were identified in July 2006, August 2008 and July 2015. These represent short-lived deviations from the normal pattern. For instance, the TC in July 2006 adds a sharp temporary increase of 1440 GWh, while the TC in August 2008 and July 2015 indicate negative and positive deviations respectively. These events may be related to transient anomalies such as extreme temperatures, power outages, or temporary economic events.

Re-estimation on adjusted series

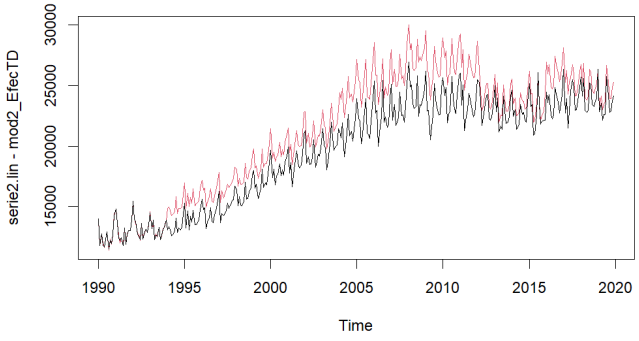
Following the removal of outliers and adjustment for calendar effects, the SARIMAX models were re-estimated using the corrected series. The new models incorporated the same orders as the previously selected ARIMA models, but now with external regressors and linearized series:

$$Y_t^{\text{adj}} = \mu + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i} + \sum_{j=1}^{Q} \Theta_j \varepsilon_{t-12j} + \beta_1 \cdot \text{TD}_t + \beta_2 \cdot \text{EAST}_t + \varepsilon_t$$

The linearized adjusted models can be observed in Figure 30.



(a) Linearized adjusted model for ARIMA(2,0,1)(0,1,1)$_{12}$



(b) Linearized adjusted model for ARIMA(1,0,2)(0,1,1)$_{12}$

Figure 30. Linearized models adjusted for calendar effects and outliers for two selected ARIMA models.

3.2. Validation

Both SARIMAX models now include the regressors modelling the trading day and Easter days. Table 12 proofs that with the inclusion of this additional information Model 1 and Model 2 all parameter estimates are still statistically significant at the 5% level, as indicated by their 95% confidence intervals not containing zero. This holds true especially for the calendar effects (trading days and Easter) indicating that their inclusion makes sense for both models and leads to an improved fit, rather than simply introducing additional noise.

Table 12. SARIMA model coefficient estimates with standard deviations and 95% confidence intervals

Model 1: SARIMA (2,0,1)(0,1,1)[12] + TD + EAST				
Param.	**Estimate**	**Std. Error**	**95% CI**	
ar1	1.3510	0.0999	1.1551	1.5469
ar2	-0.3561	0.0985	-0.5491	-0.1630
ma1	-0.6856	0.0749	-0.8325	-0.5387
sma1	-0.7334	0.0393	-0.8104	-0.6564
wTradDays	27.6189	7.1694	13.5669	41.6709
wEast	-298.7473	109.6511	-513.6634	-83.8312
Model 2: SARIMA (1,0,2)(0,1,1)[12] + TD + EAST				
Param.	**Estimate**	**Std. Error**	**95% CI**	
ar1	0.9906	0.0071	0.9767	1.0045
ma1	-0.2997	0.0521	-0.4017	-0.1976
ma2	-0.1653	0.0491	-0.2616	-0.0690
sma1	-0.7281	0.0407	-0.8079	-0.6484
wTradDays	27.5841	7.1390	13.5916	41.5765
wEast	-299.9398	106.2292	-508.1491	-91.7305

3.2.1. ACF and PACF Diagnostics

We compare the theoretical ACF and PACF from both models the with the ACF and PACF linearized, displayed in Figure 31 and Figure 32. Again the sample and theoretical ACF and PACF coincide almost perfectly. All significant lags that are visible in the sample ACF and PACF plot follow their theoretical counterpart very closely. For the ACFs of both models the decaying pattern in the ACF and the AR/SAR lags in the PACF are clearly represented. This indicates that the additional regressors and outlier treatment did not invalidate the tow models and rather contributed in cleaning and explaining the autocorrelation structure of the series.

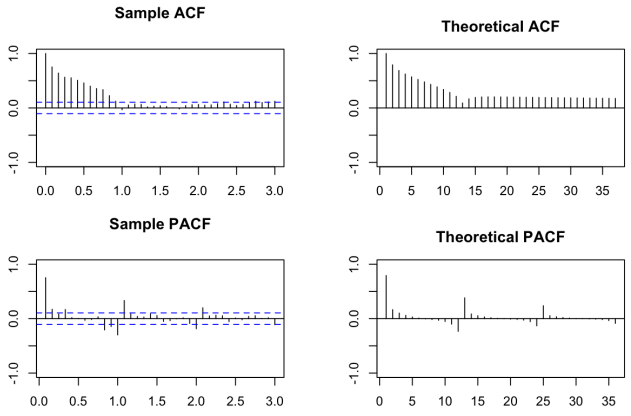


Figure 31. Theoretical and Sample ACF and PACF of ARIMAX(2,0,1)(0,1,1).

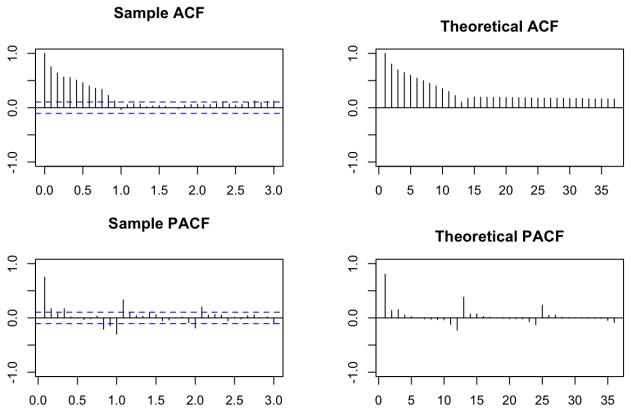


Figure 32. Theoretical and Sample ACF and PACF of ARIMAX(1,0,2)(0,1,1).

3.2.2. Residual Assumptions

In this section, we assess whether the residuals of both models still can be considered to meet the assumption of (Gaussian) white noise. Again, at first sight, the residual distributions analyzed in Figure 34 and Figure 34 still seem to satisfy this condition for both models at least approximately. However, the deviations from the normal distribution seem to have slightly increased. We continue by investigating the white noise properties quantitatively using the statistical tests introduced in the ARIMAX validation.

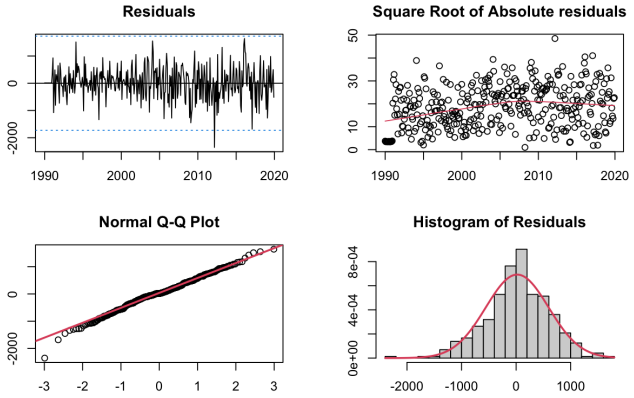


Figure 33. Residual Overview for ARIMAX(2,0,1)(0,1,1)

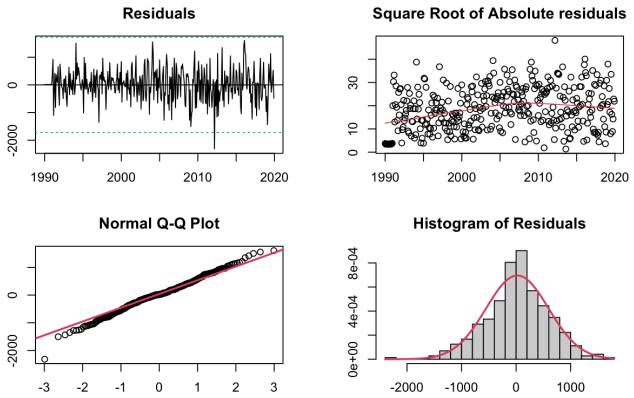


Figure 34. Residual Overview for ARIMAX(1,0,2)(0,1,1)

1. Zero Mean To evaluate whether the residuals of the two ARIMAX models have a mean of zero, we conduct one-sample *t*-tests. The test results are summarized in Table 13.

Both ARIMAX(2,0,1)(0,1,1) and ARIMAX(1,0,2)(0,1,1) models yield p-values of 0.5108 and 0.4689, respectively. Since both p-values are well above the conventional significance level of 0.05, we fail to reject the null hypothesis in both cases.

This indicates that there is no statistically significant evidence to suggest that the residuals deviate from a zero mean. Therefore, we conclude that the residuals of both ARIMAX models can be considered to have zero mean, which satisfies the first key assumption of white noise residuals.

Table 13. One-sample t-test results for zero-mean residuals

Statistic	ARIMAX(2,0,1)(0,1,1)	ARIMAX(1,0,2)(0,1,1)
Mean (Sd.)	19.9824 (576.0335)	21.9126 (573.4285)
t-Statistic	0.6582	0.7251
p-Value	0.5108	0.4689
95% CI	[−39.7226, 79.6875]	[−37.5224, 81.3477]

2. Homoskedasticity To assess whether the residuals of the ARIMAX models exhibit constant variance (homoskedasticity), we conducted the Breusch–Pagan test. The null hypothesis of the test is that the residuals are homoskedastic. The test results are reported in Table 14.

For both ARIMAX(2,0,1)(0,1,1) and ARIMAX(1,0,2)(0,1,1), the Breusch–Pagan statistics are 18.822 and 17.131, respectively, with associated p-values smaller than 0.0001. Since these p-values are well below the 0.05 significance threshold, we reject the null hypothesis for both models.

This indicates strong evidence of heteroskedasticity in the residuals of both ARIMAX models. As a result, the assumption of constant variance is violated, which may affect the reliability of inference based on these models.

To address this issue, one possible remedy is to use heteroskedasticity-consistent standard errors when estimating parameters and constructing confidence intervals. Alternatively, a more comprehensive solution is to explicitly model the conditional variance using a GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model. As these approaches are beyond the scope of this project, we have to treat variance estimates and confidence intervals in the following with caution.

Table 14. Breusch–Pagan test for homoskedasticity

Statistic	ARIMAX(2,0,1)(0,1,1)	ARIMAX(1,0,2)(0,1,1)
BP statistic	18.822	17.131
DF	1	1
p-value	<0.0001	<0.0001

3. Independence

To verify the assumption of independent residuals, we applied the Durbin–Watson (DW) test and the Ljung–Box test at multiple lags. The results are presented in Tables 15 and 16.

The DW statistics for both models are close to 2 (1.9734 for ARIMAX(2,0,1)(0,1,1) and 1.9655 for ARIMAX(1,0,2)(0,1,1)), and their corresponding p-values (0.3799 and 0.3513) are not statistically significant. This indicates no evidence of first-order autocorrelation in the residuals.

Table 15. Durbin-Watson test results for both models. The DW statistics near 2 and the non-significant p-values indicate no first-order autocorrelation.

Model	DW Statistic	p-value
ARIMAX(2,0,1)(0,1,1)	1.9734	0.3799
ARIMAX(1,0,2)(0,1,1)	1.9655	0.3513

Furthermore, the Ljung–Box test conducted at lags 12, 24, 36, and 48 yields high p-values for both models across all lags, suggesting the absence of significant autocorrelation at higher orders as well.

These results provide strong support for the independence assumption of the residuals in both ARIMAX models.

Table 16. Ljung-Box test results at multiple lags for both models. The high p-values indicate no significant autocorrelation in the residuals.

Model	Lag	Statistic	p-value
ARIMAX(2,0,1)(0,1,1)	12	10.3549	0.5849
	24	22.2424	0.5648
	36	27.4818	0.8451
	48	46.2925	0.5430
ARIMAX(1,0,2)(0,1,1)	12	10.5337	0.5692
	24	22.0134	0.5785
	36	26.7420	0.8689
	48	44.7428	0.6071

4. Normality To assess whether the residuals of the SARIMAX models are normally distributed, we again perform three statistical tests: the Shapiro–Wilk test, the Anderson–Darling test, and the Jarque–Bera test. The results are reported in Table 17.

For both models, the majority of the tests reject the null hypothesis of normally distributed residuals at the 5% significance level. In particular, the Jarque–Bera test yields low p-values (0.0072 for ARIMAX(2,0,1)(0,1,1) and 0.0276 for ARIMAX(1,0,2)(0,1,1)), indicating deviations from normality in skewness and kurtosis. Similarly, the Anderson–Darling test shows significant results for both models.

While the Shapiro–Wilk test indicates marginal significance for ARIMAX(2,0,1)(0,1,1) (with $p = 0.0402$) and borderline insignificance for ARIMAX(1,0,2)(0,1,1) (with $p = 0.0681$), the overall pattern suggests non-normality of residuals.

These results imply that the assumption of normality is not fully satisfied for either model. Although this does not necessarily invalidate the models' predictive performance, it suggests caution when applying inferential procedures that rely on normally distributed residuals. In future work, modeling residuals with non-Gaussian distributions or using robust statistical methods may be considered.

Table 17. Normality test results for residuals of both SARIMAX models

Test	Model	Statistic	p-value
Shapiro-Wilk	ARIMAX(2,0,1)(0,1,1)	0.9917	0.0402
	ARIMAX(1,0,2)(0,1,1)	0.9925	0.0681
Anderson-Darling	ARIMAX(2,0,1)(0,1,1)	0.9260	0.0185
	ARIMAX(1,0,2)(0,1,1)	0.8516	0.0283
Jarque-Bera	ARIMAX(2,0,1)(0,1,1)	9.8782	0.0072
	ARIMAX(1,0,2)(0,1,1)	7.1795	0.0276

3.2.3. Stationarity (Causality) and Invertibility

The ARIMAX models extend the previously analyzed SARIMA models by incorporating trading days and Easter effects as external regressors and applying outlier treatment. The functional form is otherwise unchanged.

The stationarity (or causality) of the models depends on the AR polynomial roots. For the extended ARIMA(2,0,1)(0,1,1) model with regressors, the moduli of the AR roots are 1.008 and 2.786, both greater than 1. Similarly, the extended ARIMA(1,0,2)(0,1,1) model has an AR root modulus of 2.819. Thus, both models are stationary.

π-weights, which reflect the influence of past observations in the AR(∞) representation, are visualized in Figures 35 and 36. For both models, the weights decay rapidly over the first 10 lags and remain

small thereafter. Minor seasonal fluctuations appear around lag 12 due to monthly seasonality but do not dominate, confirming weak long-term dependence and stationarity.

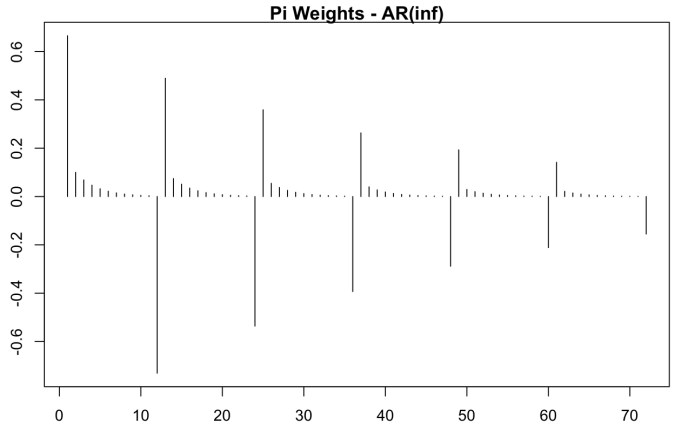


Figure 35. π-weights of ARIMAX(2,0,1)(0,1,1).

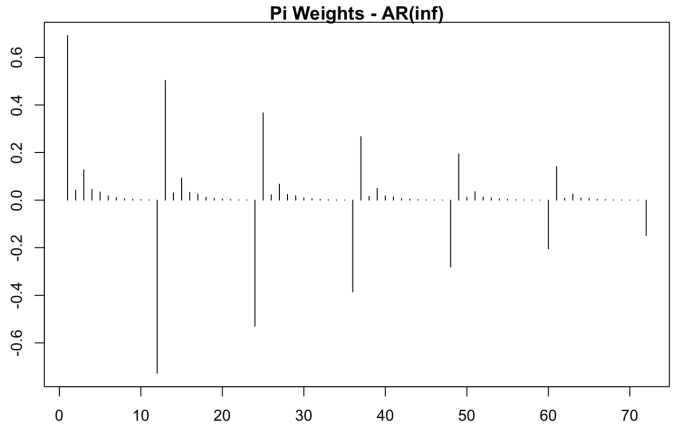


Figure 36. π-weights of ARIMAX(1,0,2)(0,1,1).

For invertibility, we check the roots of the MA polynomial. All moduli for both models are greater than 1, with the smallest around 1.026 and the largest at 4.979, confirming invertibility.

The ψ-**weights** shown in Figures 37 and 38 decay over time, as expected. The values decrease sharply from approximately 0.66 to below 0.08 by lag 20. Although there are slight fluctuations around lag 12—likely due to seasonal effects—the overall trend supports invertibility.

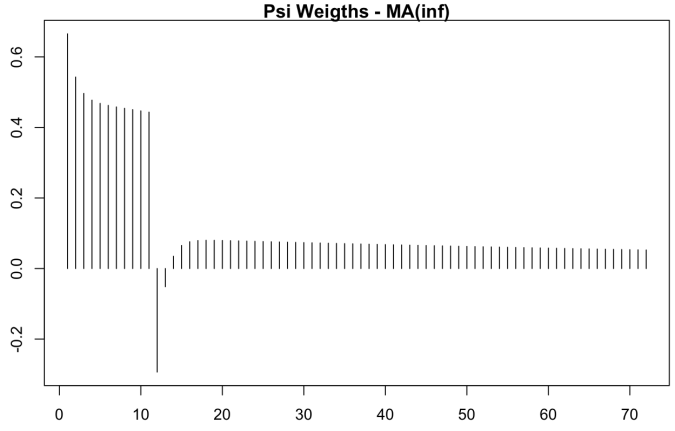


Figure 37. ψ-weights of ARIMAX(2,0,1)(0,1,1).

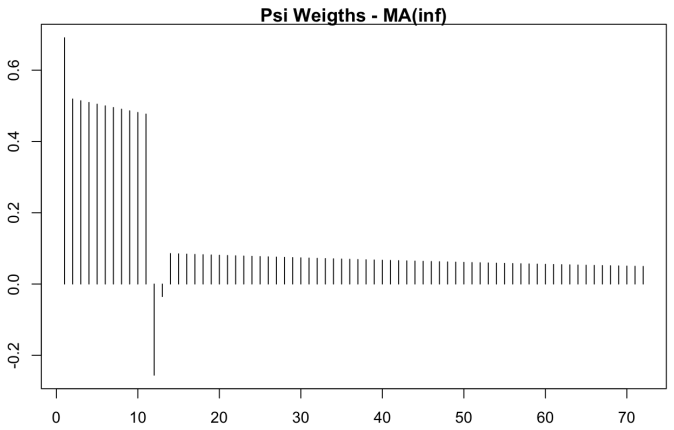
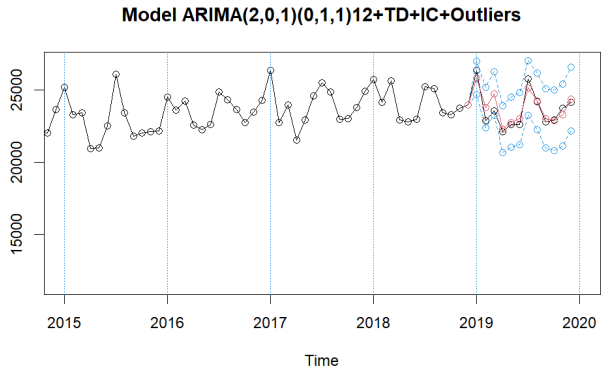


Figure 38. ψ-weights of ARIMAX(1,0,2)(0,1,1).

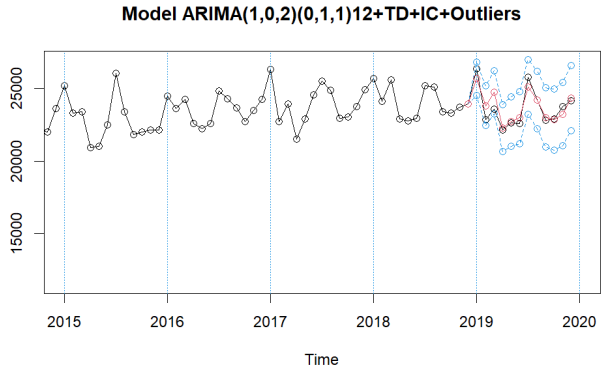
In conclusion, based on AR and MA root moduli as well as the behavior of the π- and ψ-weights, **both ARIMAX models meet the conditions for stationarity (causality) and invertibility**. The addition of trading day and Easter effects did not alter the fundamental dynamics of the original SARIMA structures. The outlier treatment also did not alter the series in a way that the models became invalid.

3.3. Prediction

As for the ARIMA models to assess the predictive performance and stability of the selected models, we reserved the last 12 months of the series for out-of-sample validation. Using the estimated SARIMAX models, we generated one-step-ahead forecasts and compared them to the actual observed values. The predictions can be observed in Figures 39.



(a) Prediction with ARIMA(2,0,1)(0,1,1) adjusted for calendar effects and outliers$_{12}$



(b) Prediction with ARIMA(1,0,2)(0,1,1) adjusted for calendar effects and outliers$_{12}$

Figure 39. Stability and one-year predictions for the two selected ARIMAX models, showing the observed series, predicted values, and 95% confidence intervals.

As before error metrics RMSE, MAE, RMPSE and MAPE were calculated and can be observed in Table 18.

Model	RMSE	MAE	RMSPE	MAPE
ARIMAX(2,0,1)(0,1,1)$_{12}$	537.08	410.80	0.02262	0.01726
ARIMAX(1,0,2)(0,1,1)$_{12}$	558.31	423.39	0.02342	0.01774

Table 18. Forecast error metrics for the two ARIMAX models with calendar effects and outlier adjustment.

3.4. Forecasting

Once the ARIMAX models were estimated using the linearized series (adjusted for calendar effects and outliers), forecasts were generated for the 12 months following December 2019. The goal was to evaluate whether incorporating variables and correcting structural anomalies improves forecast accuracy and precision.

Forecast computation

Forecasts were produced using the `predict()` function with future values of the regressors (trading days and Easter) provided through the `newxreg` argument. The cumulative impact of identified Level Shift (LS) outliers was added to the forecasted values to account for permanent shifts in the series.

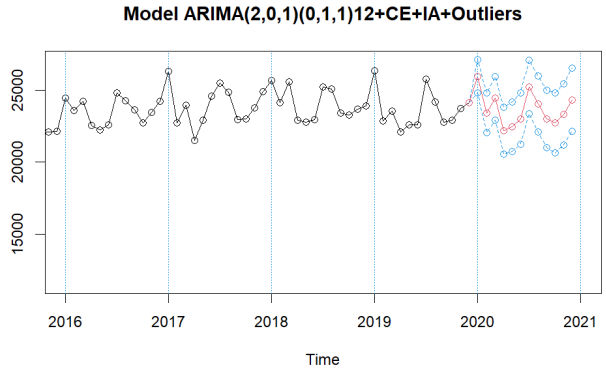
Let $\hat{y}_{T+h}$ be the h-step-ahead forecast, and $\hat{\sigma}_h$ its associated standard error. The final forecast including the LS adjustment is:

$$\hat{Y}_{T+h} = \hat{y}_{T+h} + w_{\text{LS}}$$

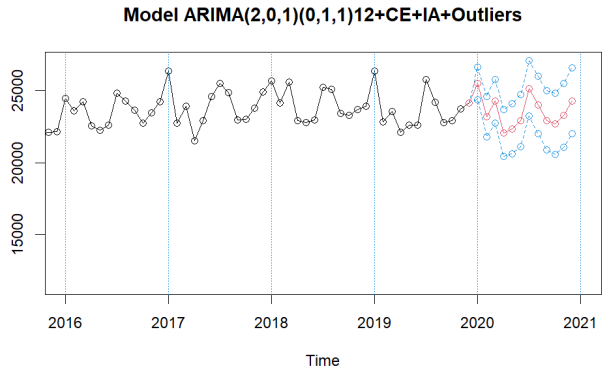
Prediction intervals were calculated as:

$$\hat{Y}_{T+h} \pm 1.96 \cdot \hat{\sigma}_h$$

The forecast can be observed in fig 40.



(a) Forecast with ARIMAX(2,0,1)(0,1,1)$_{12}$



(b) Forecast with ARIMAX(1,0,2)(0,1,1)$_{12}$

Figure 40. One-year forecast for the two selected ARIMAX models, showing the observed series, predicted values, and 95% confidence intervals.

4. C. Comparison of ARIMA and ARIMAX

In this section, we compare the performance of SARIMA models with and without exogenous variables (ARIMAX). Table 19 shows the **in-sample fit** metrics—Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and log-likelihood—for both standard SARIMA and extended SARIMAX models.

Table 19. In-sample fit comparison of SARIMA and SARIMAX models

Model	AIC	BIC	LogLik
SARIMA(2,0,1)(0,1,1)	-1431.480	-1408.164	721.740
SARIMA(1,0,2)(0,1,1)	-1430.814	-1407.497	721.407
SARIMAX(2,0,1)(0,1,1)	5446.75	5473.72	-2716.38
SARIMAX(1,0,2)(0,1,1)	5443.56	5470.526	-2714.78

The standard SARIMA models (without exogenous regressors) exhibit significantly lower AIC and BIC values compared to their

ARIMAX counterparts. However, this is expected, as the SARIMAX models were trained on a different scale and data transformation due to the outlier treatment. Within each model class, the ARIMA(2,0,1)(0,1,1)$_{12}$ slightly outperforms ARIMA(1,0,2)(0,1,1)$_{12}$ in terms of AIC and log-likelihood, although both are nearly equivalent in in-sample fit.

Table 20. Forecast error metrics for SARIMA and SARIMAX models over the 2019 validation period

Model	RMSE	MAE	RMSPE	MAPE
SARIMA(2,0,1)(0,1,1)	646.03	526.66	0.02779	0.02253
SARIMA(1,0,2)(0,1,1)	646.67	524.91	0.02783	0.02246
SARIMAX(2,0,1)(0,1,1)	537.08	410.80	0.02262	0.01726
SARIMAX(1,0,2)(0,1,1)	558.31	423.39	0.02342	0.01774

Table 20 reports the **out-of-sample forecast performance** for all models. The ARIMAX models clearly outperform the plain SARIMA models across all error metrics. Specifically, the ARIMAX(2,0,1)(0,1,1)$_{12}$ model achieves the lowest root mean squared error (RMSE) and mean absolute error (MAE), indicating improved predictive accuracy after accounting for calendar effects and external regressors.

In conclusion, the ARIMAX(2,0,1)(0,1,1)$_{12}$ model with trading day and Easter effects delivers the best overall performance, both in terms of in-sample fit and forecast accuracy, making it the preferred model among the evaluated candidates.

5. Conclusion

This project examined the applicability of SARIMA and SARIMAX models for forecasting Gross Internal Electrical Energy Consumption in Spain from 1990 to 2020. By incorporating calendar-related exogenous variables—specifically trading days and Easter effects—we demonstrated that the SARIMAX models consistently outperform their SARIMA counterparts in both in-sample fit and out-of-sample forecast accuracy.

5.1. Key Findings

The inclusion of exogenous variables in the SARIMAX framework led to significant improvements in model performance. This is evident from the notable reduction in RMSE and MAE values over the 2019 validation period. The best-performing model, SARIMAX(2,0,1)(0,1,1)$_{12}$, effectively captured the seasonality and calendar-related fluctuations in electricity consumption.

Outlier treatment and the careful construction of trading day and Easter indicators further enhanced model robustness. These results highlight the importance of incorporating structural and contextual knowledge into statistical time series models, especially in domains with known calendar or policy-related disruptions.

5.2. Limitations and Outlook

Despite the success of the SARIMAX approach, several limitations should be acknowledged:

- **Limited exogenous variable set:** Only Easter and trading days were considered. Other recurring holidays and socio-economic factors—such as public holidays, weather, or major events—may also significantly influence electricity consumption but were not modeled.
- **Assumed normality:** Diagnostic checks and model selection criteria implicitly assume normally distributed errors, which may not fully capture the true error distribution, potentially leading to distorted test results. Mostly, applying the Central Limit Theorem can resolve this, but still leads to potentially underestimated residual variances. For this reason results from inference have to be treated with caution.
- **Assumed noise-free structure:** SARIMA and SARIMAX models treat the observed process as clean aside from the modeled noise. For the SARIMA models this assumption could be strongly approved. However, for the SARIMAX models there signs of heteroskedasticity and non-normality became apparent. Extensions using state space models or stochastic volatility approaches (e.g., GARCH-type models) could better reflect dynamic volatility and uncertainty.
- **Sensitivity to structural changes:** The electricity consumption series is susceptible to sudden changes due to policy shifts or structural transformations in the energy mix (e.g., increased share of renewables), which for future forecasts cannot be accounted for in static SARIMAX models, even with in-sample outlier treatment.

5.3. Final Remark

The results confirm that ARIMAX-type models can provide reliable and interpretable forecasts for electricity demand. Their ability to reflect underlying trends and seasonality, while incorporating meaningful external information, makes them particularly useful for operational and planning purposes in the energy sector.

Overall, this study demonstrates that while classical SARIMA models provide a solid baseline, SARIMAX models—especially when enhanced with relevant calendar features and outlier adjustments—offer a more accurate and context-aware forecasting tool for electricity consumption in Spain. The approach is especially well-suited for modeling structured seasonal patterns and is robust enough for short- to medium-term forecasting applications.

References

[1] A. Jahanshahi, D. Jahanianfard, A. Mostafaie, and M. Kamali, "An auto regressive integrated moving average (arima) model for prediction of energy consumption by household sector in euro area", *AIMS Energy*, vol. 7, no. 2, pp. 151–164, 2019. DOI: 10.3934/energy.2019.2.151.

[2] F. R. Alharbi and D. Csala, "A seasonal autoregressive integrated moving average with exogenous factors (sarimax) forecasting model-based time series approach", *Inventions*, vol. 7, no. 4, p. 94, 2022. DOI: 10.3390/inventions7040094.

[3] A. D. Morgoeva, O. A. Gavrina, N. V. Martyushev, E. A. Efremenkov, and Q. Mengxu, "Methods of forecasting electric energy consumption: A literature review", *Energies*, vol. 15, no. 23, p. 8919, 2022. DOI: 10.3390/en15238919.

[4] N. Maleki, O. Lundström, A. Musaddiq, J. Jeansson, T. Olsson, and F. Ahlgren, "Future energy insights: Time-series and deep learning models for city load forecasting", *SSRN Electron. J.*, 2024. DOI: 10.2139/ssrn.4811409.

6. Appendix

Table A1: Modulus of AR and MA Roots for Model 1 (SARIMA(2,0,1)(0,1,1))

AR Roots (Stationarity)	MA Roots (Invertibility)
1.005305	1.019923
4.327857	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.019923
	1.236331

Table 21. Modulus of AR and MA roots for Model 1. The AR roots suggest stationarity; the MA roots indicate invertibility.

Table A2: Modulus of AR and MA Roots for Model 2 (SARIMA(1,0,2)(0,1,1))

AR Roots (Stationarity)	MA Roots (Invertibility)
1.005522	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.019957
	1.269747
	4.979622

Table 22. Modulus of AR and MA roots for Model 2. The AR part is stationary; the MA part is invertible.

Table A3: Modulus of AR and MA Roots for ARIMAX(2,0,1)(0,1,1)

AR Roots (Stationarity)	MA Roots (Invertibility)
1.008000	1.026177
2.786058	1.026177
	1.026177
	1.026177
	1.026177
	1.026177
	1.026177
	1.026177
	1.026177
	1.026177
	1.026177
	1.026177
	1.458525

Table 23. Modulus of AR and MA roots for ARIMAX(2,0,1)(0,1,1). The AR roots indicate stationarity; the MA roots suggest invertibility.

Table A4: Modulus of AR and MA Roots for ARIMAX(1,0,2)(0,1,1)

AR Roots (Stationarity)	MA Roots (Invertibility)
1.009485	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.026795
	1.714917
	3.527799

Table 24. Modulus of AR and MA roots for ARIMAX(1,0,2)(0,1,1). The AR part is stationary; the MA part is invertible.